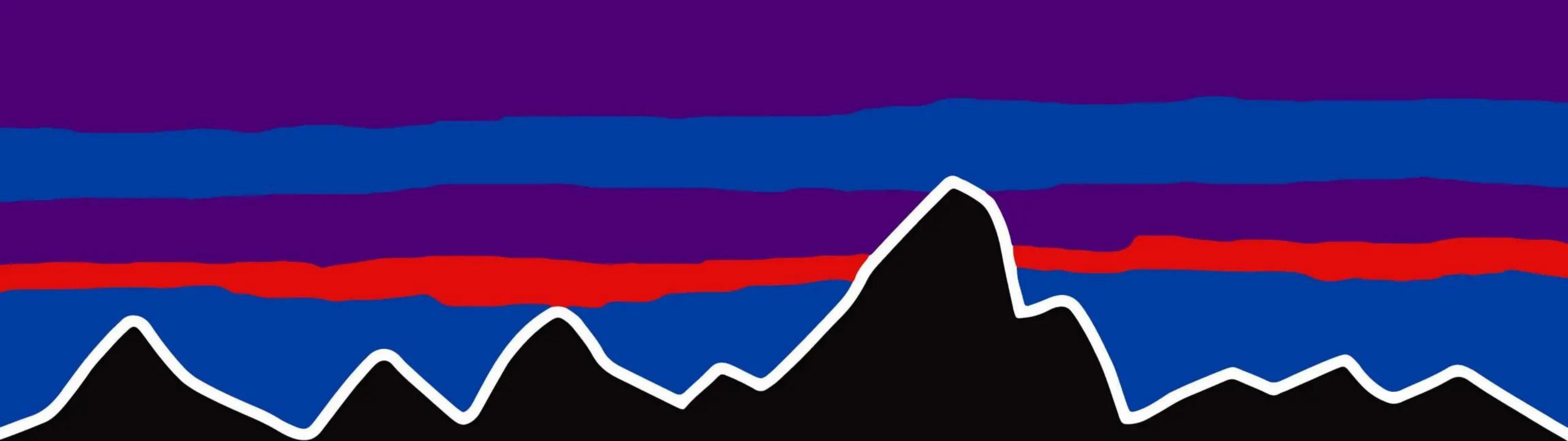




The supply chain
of

patagonia[®]

By Vinayak Singh



What is Patagonia?

Patagonia is an American retailer of [outdoor recreation](#) clothing. It was founded by [Yvon Chouinard](#) in 1973 and is based in [Ventura, California](#). Patagonia operates stores in more than 10 countries globally, as well as factories in 16 countries.

In 2023, Patagonia generated revenue of 1.5 billion euros.

Patagonia has been committed for 50 years to donating 1% of its sales to environmental organizations.

Patagonia is often the subject of case studies regarding sustainable and innovative business models. It notably promotes transparency in its operations, which is why we have chosen this company to analyze its supply chain.

Patagonia's value chain stands out for its integration of sustainability and ethical practices at every step, from product design to delivery to the end consumer. This approach not only minimizes environmental impact but also reinforces the company's commitment to social responsibility.

The company places a high priority on sourcing materials in ways that are environmentally friendly and socially responsible. This includes using organic cotton, recycled polyester, and ensuring that all animal-derived products (like wool and down) are sourced from operations that are humane and sustainable.

The Manufacturing process(1/3)

01

The design and pattern making (1-2 months)**

According to Patagonia, "90% of a product's environmental impact is determined at the design stage". The design step is therefore one of the most time consuming steps of the manufacturing process. Designers create the down sweater's blueprint, considering factors like warmth, weight, and weatherproofing

02

Raw materials acquisition (2 weeks)

Patagonia selects carefully the raw materials used for its products. For this jacket, 2 main materials are used : recycled nylon and responsible down. In order to produce this jacket, the raw material need to be purchased and sent to the facilities. The duration of this step depends on the supplier's location.



**Patagonia sells many different kind of products, like sweateer, shoes and even bags.

In order to analyse the manufacturing process, we will focus on one of Patagonia's best seller : **The Down Sweater jacket.**



The Manufacturing process(2/3)

03

Material preparation (1-2 weeks)

Raw materials undergo various preparation and processing steps to make them suitable for manufacturing. For this jacket, the nylon has to be coated with a water repellent finish for example.

04

Fabric Cutting (1 week)

Once the final design is approved, fabric for the shell, liner, and any additional features (pockets, hood) are precisely cut using large cutting machines or lasers.

05

Assembly (2-4 weeks)

Skilled workers sew together the various pieces of fabric, incorporating zippers, buttons, and any special features. The down is carefully measured and loaded into the shell. Specialized machines may distribute the down evenly to create consistent warmth.



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The Manufacturing process(3/3)

06

Quality Control (1-2 weeks)

Raw materials undergo various preparation and processing steps to make them suitable for manufacturing. For this jacket, the nylon has to be coated with a water repellent finish for example.

07

Product testing (2 weeks)

Testing the product in real conditions is an important part of the manufacturing process of Patagonia. They want to make sure that the product will be resistant. The duration is dependent on the supplier's location.

08

Packaging and Shipping (1-3 weeks)

Once the down sweater is manufactured and passes quality control checks, it is packaged and shipped to distributors or retail stores. This can take weeks or months depending on location.



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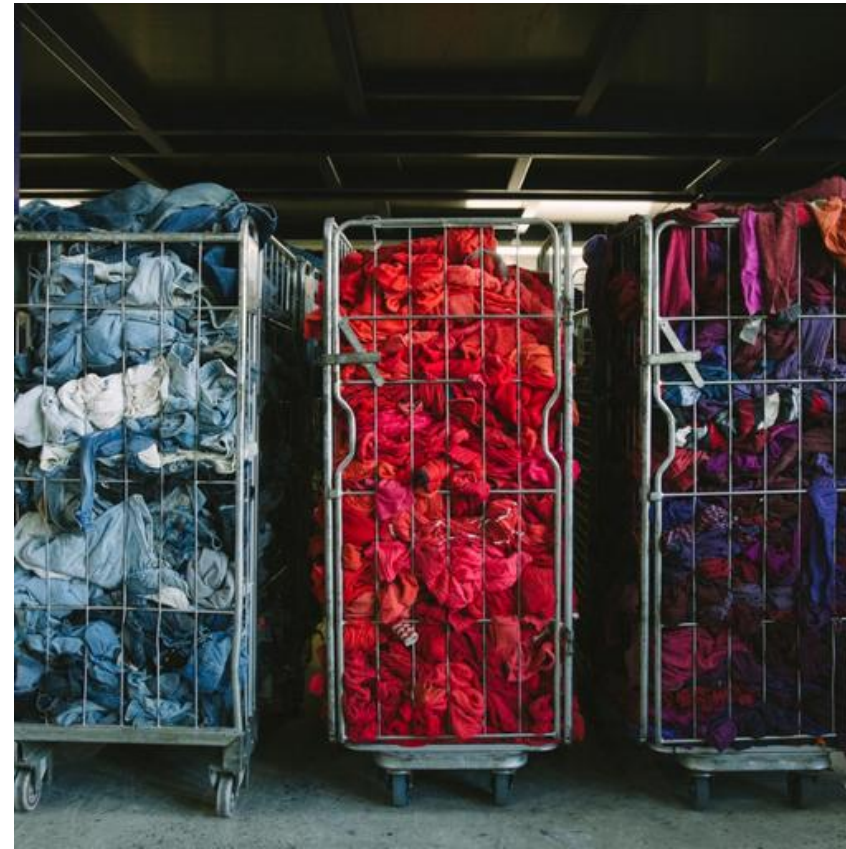
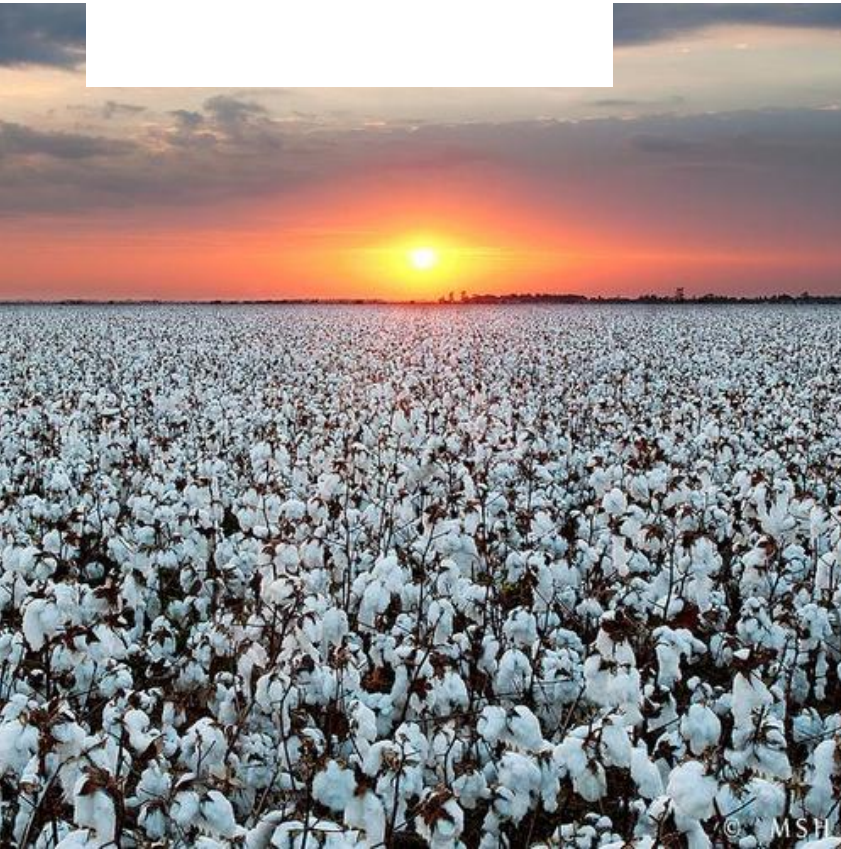
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Supply Chain



Patagonia's supply chain isn't just a thread—it's a tapestry woven with sustainability, ethics, and innovation. From traceable materials to pioneering programs like the Supply Chain Environmental Responsibility and Raw Materials Social Responsibility initiatives, Patagonia stitches together a narrative of responsible business, where every fiber tells a story of conscious consumption and environmental stewardship.



Key Partners

01

Raw material suppliers:

- Organic cotton farms
- Recycled material processors*
- Responsible Down Standard (RDS) certified farms**

02

Fabric mills and production factories :

These facilities located worldwide have skilled workers who cut fabric pieces, assemble garments, and ensure quality control following Patagonia's specifications. Patagonia currently works with 64 of these partners, the list of which is available on their website.

03

Logistics Providers

Companies that handle the transportation of raw materials, finished goods, and products between different stages of the supply chain.

04

Distributors and Retailers

Wholesalers and retail stores that purchase Patagonia products in bulk and sell them to consumers.

*Partners who specialize in transforming post-consumer or pre-consumer waste into recycled materials like polyester or nylon used in Patagonia's products.

**Suppliers who raise geese for food and humanely collect the down feathers for Patagonia's Traceable Down Standard.

On our previous example of the Down Jacket, Patagonia works with ECONYL, which is a partner that is able to treat recycled nylon to give it back its original characteristics.



The bottleneck, throughput time, cycle time, and throughput rate when we order the Dawn Sweater Jacket.



I have voluntarily excluded the jacket design activities and the quality control of the final product because we consider these activities only relevant to the pre-commercialization phase of the jacket and therefore, they do not affect the production process once the customer's order has been placed.



The bottleneck, throughput time, cycle time, and throughput rate when we order the Dawn Sweater Jacket.

- **Bottleneck** (the activity with the largest processing time or lowest output) = Assembly of up to 4 weeks
- **Throughput time** (the sum of processing time for all activities) = $2+2+1+4+2+3 = 14$ weeks
- **Cycle time** (The rate at which products comes out of the system or the process time of bottleneck activity) = 4 weeks
- **Throughput rate** (the rate at which the system produces one unit) = it takes roughly 672 hours to produce a single jacket when considering the bottleneck activity, which is the Assembly stage with a cycle time of 4 weeks.



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How does Patagonia manage its finished goods and raw materials ?(1/2)

Handling

Patagonia manages its finished goods and raw materials through an approach that emphasizes social and environmental responsibility across its entire supply chain. For their finished goods factories, Patagonia engages in rigorous due diligence activities, ensuring compliance with fair labor practices, safe working conditions, and environmental standards. This involves a comprehensive Supplier Workplace Code of Conduct that is aligned with the International Labour Organization's core labor standards and covers everything from labor practices to health and safety, environmental impact, and community engagement.

Focus on the 4-Fold Approach to Supply Chain Decisions:

The company has a 4-Fold Approach to Supply Chain Decisions to pre-screen factories before placing orders, requiring suppliers to meet their business, quality, social, and environmental criteria. This approach ensures that every new and existing supplier aligns with Patagonia's high standards, and their teams have the power to veto any new factory that doesn't meet these standards.





How does Patagonia manage its finished goods and raw materials ?(2/2)

Raw Materials

For raw materials, Patagonia has a robust Material Traceability program to ensure a strong chain of custody and is moving towards using recycled materials and regenerative organic sources for its products. They are actively engaged in measuring and reducing the environmental impacts of manufacturing Patagonia products, emphasizing the need for suppliers to comply with local laws and Patagonia's own code of conduct. All of these elements ensure transparency and accountability throughout the supply chain

Green Commitment

Patagonia also has an ongoing commitment to reduce its carbon footprint and has set a goal to use preferred materials, like organic and regenerative organic cotton, recycled polyester, and recycled nylon, by 2025, thus minimizing their dependence on new raw materials and reducing carbon emissions. Patagonia's commitment to using recycled materials reduces the need for virgin resources, further lowering their carbon footprint





How does Patagonia prepare its production plans ? What data are used by them ?(1/2)

Planning

Patagonia's approach planning process integrates rigorous data analysis, advanced forecasting techniques, and stakeholder engagement to align with its commitment to sustainability and social responsibility. They are driven by a philosophy that prioritizes quality and minimizing harm over mere expansion and profit. This approach is infused into every aspect of their operations, including production planning, and hinges on a profound understanding of their environmental impact and social responsibility.

Carbon footprint

A significant part of Patagonia's production planning involves assessing the carbon footprint across its supply chain, which is considerable, as nearly 86% of the company's total emissions come from material creation used in their products. To reduce this footprint and move towards their goal of carbon neutrality by 2025, Patagonia implements multiple strategies that focus on the materials they source and the processes involved in creating their apparel. This includes sourcing sustainable materials, optimizing transportation logistics to minimize emissions, and investing in renewable energy solutions for their facilities



The Size of the Company 'Given Away' to Save the Planet

Estimated global revenue of Patagonia, Inc.
(in million U.S. dollars)



Source: Statista Company DB





How does Patagonia prepare its production plans ? What data are used by them ?(2/2)

Monitoring Supplier Plans

Patagonia has also embraced a more transparent and ethical approach by working closely with suppliers that share their values and committing to fair labor practices. This includes rigorous screening processes to ensure that factories and mills meet their environmental and social standards. As such, when preparing production plans, they likely consider the compliance of their suppliers with their established Supplier Workplace Code of Conduct, which covers a wide range of aspects from labor and safety to environmental impact and community engagement while serving as a benchmark for evaluating compliance across various aspects

Data Collected

On top of everything mentioned earlier Patagonia uses the below data as well for making production plans :

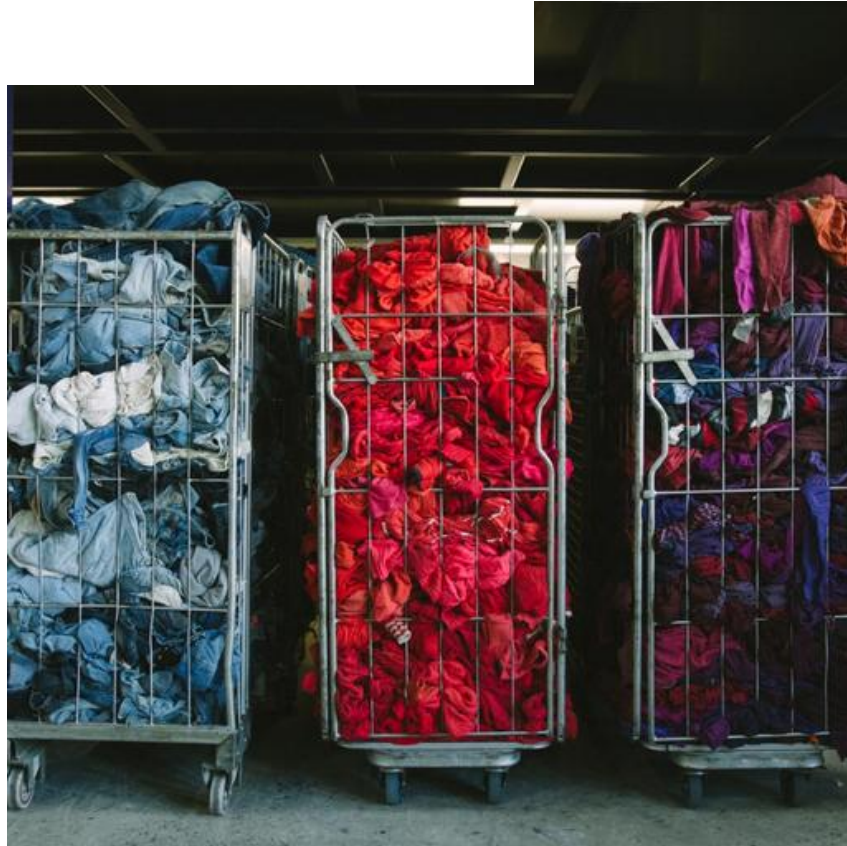
- **Sales data:** This includes historical sales figures for different product categories, regions, and seasons. This helps them understand what's selling well and plan production accordingly and guiding production decisions.
- **Inventory data:** Knowing current inventory levels of finished goods, raw materials, and work-in-progress helps avoid overproduction or stockouts and optimize inventory management.
- **Material lead times:** Understanding how long it takes to receive raw materials from suppliers helps determine precise production timelines.
- **Production capacity:** Knowing the limitations of their own factories or those of their suppliers helps plan realistic production volumes.
- **Seasonality:** Patagonia is good at making plans for increased production of winter gear in advance of the winter season and vice versa for summer.





How does the company identify the various material requirements?(1/2)

Finding completely harmless materials remains a challenge (e.g., DWR coatings). Patagonia acknowledges these issues and seeks continuous improvement by employing a comprehensive strategy for material selection, prioritizing functionality, environmental responsibility, and ethical sourcing practices.



01 Functional Need and Environmental Assessment:

- Patagonia employs a comprehensive approach to material selection, considering both functional necessity and environmental impact.
- The company's decisions are guided by environmental profit and loss metrics, ensuring that product selection aligns with minimizing socio-ecological harm.

03 Responsible Sourcing:

- Patagonia prioritizes responsible sourcing practices, seeking raw materials with minimal socio-ecological impacts, even at a potential financial loss.
- Initiatives such as the transition to 100% organic cotton demonstrate a commitment to environmentally friendly materials.

02 Design Stage Impact:

- sustainable design principles.
- Design criteria include repairability, ease of care, and timelessness, aiming to extend product lifespan and reduce consumption.



How does the company identify the various material requirements?(2/2)

04

Recycled and Environmentally Friendly Materials

- The company aims to increase the use of recycled and environmentally friendly materials, with a goal to utilize only recycled petroleum-based fibers by 2025.
- Pre-cycling programs further reduce virgin material usage, exemplified by the creation of infant clothing from scrap materials.

05

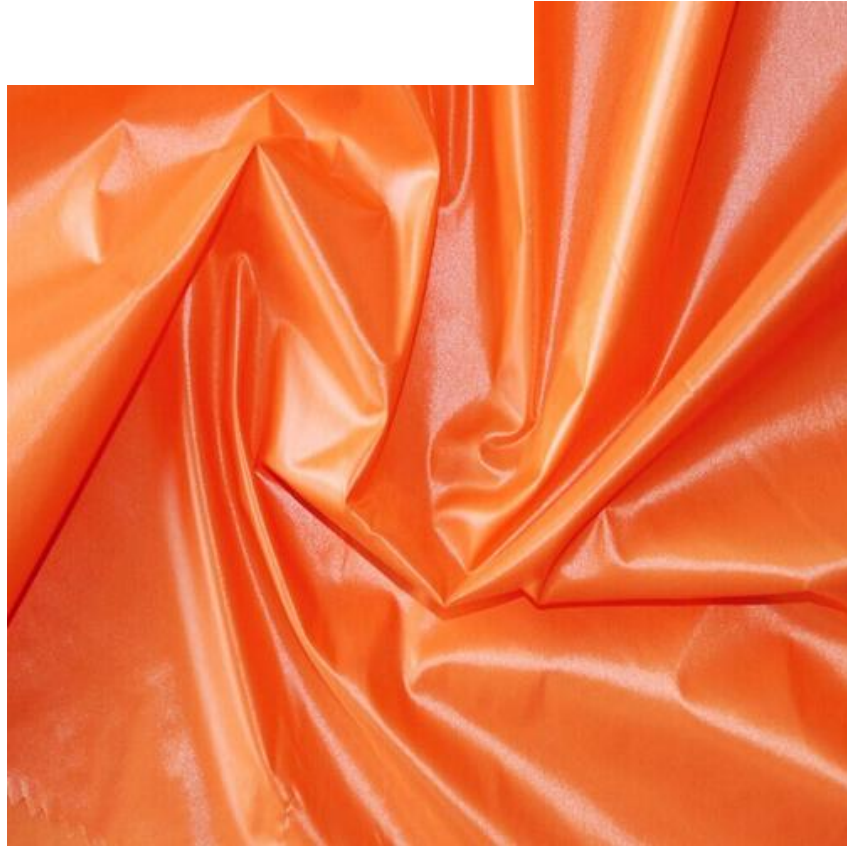
Ethical Standards and Traceability

- Patagonia implements stringent standards for animal-related inputs and ensures traceability through initiatives like the Traceable Down Standard.
- Socially responsible practices extend to monitoring working conditions at suppliers, including second or third-tier suppliers.

06

Challenges and Ongoing Improvement

- Despite efforts to minimize harm, challenges persist, such as the use of toxic chemicals in Durable Water Repellent (DWR) coatings.
- Patagonia acknowledges these shortcomings and works towards continuous improvement in material selection and production processes (O'Rourke & Strand, 2017).





What are the different quality issues faced? What quality management tools and techniques are used?(1/3)

Patagonia employs a comprehensive quality management system that combines expert evaluation, thorough testing, and a commitment to continuous improvement. This ensures materials meet their high standards for durability, functionality, aesthetics, and sustainability matching both its quality standards and its vision of environment preservation.

Quality Issues and Management

1. Quality Standards:

- Patagonia emphasizes stringent quality standards for its materials, demanding durability, functionality, ease of care, and aesthetic appeal.
- Every material must resist degradation from wearing and washing and be grown or manufactured with consideration for environmental and social impacts.

2. Quality Issues:

- Quality issues may include material degradation, inadequate performance, or failure to meet sustainability criteria.
- Ensuring consistent quality across a diverse range of materials sourced globally presents a significant challenge.



What are the different quality issues faced? What quality management tools and techniques are used?(2/3)

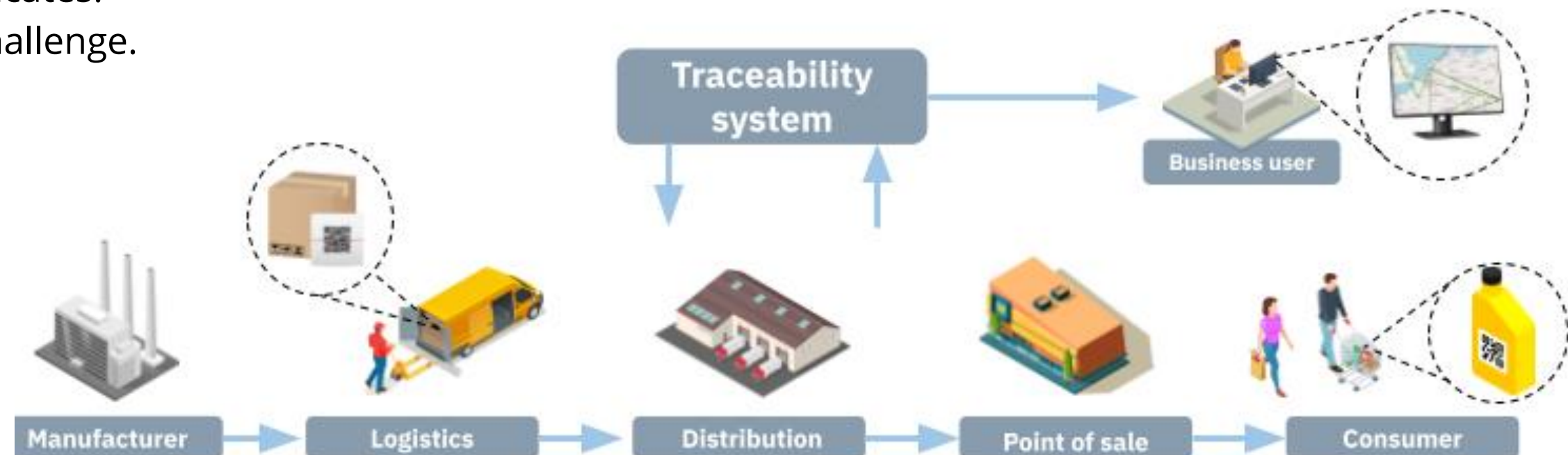
Quality Management Tools and Techniques

1. Material Testing Laboratory:

- Patagonia operates a quality testing lab where materials undergo rigorous evaluation against predefined performance criteria.
- Analysts rely on industry standards, proprietary test methods, and specialized equipment to assess material quality.

2. Traceability Requirements:

- Traceability throughout the supply chain is essential for maintaining quality and assessing environmental and social impacts.
- Suppliers are required to provide detailed information about the origin and production processes of materials, including profile sheets, supply-chain tracking sheets, and third-party certificates.
- Materials sourced globally presents a significant challenge.





What are the different quality issues faced? What quality management tools and techniques are used?(3/3)

Quality Management Tools and Techniques

1. Supply Chain Environmental Responsibility Program:

- Patagonia's Supply Chain Environmental Responsibility Program addresses environmental impacts associated with material production.
- It utilizes industry-wide tools like the Higg Index and recognizes third-party certification programs such as bluesign to ensure environmental standards are met.

2. Raw Materials Social Responsibility Program:

- The Raw Materials Social Responsibility Program monitors key material suppliers for compliance with social responsibility standards.
- This includes standards on wages, benefits, workplace safety, and other social and environmental benchmarks, ensuring ethical practices throughout the supply chain.

Vincent Stanley with Yvon Chouinard

The Future of the Responsible Company

What We've Learned from Patagonia's First 50 Years



patagonia



1.About Patagonia

Patagonia stands as a beacon of sustainability and ethical practices within the outdoor clothing industry. The company has established itself as a leader in environmentally responsible business practices

2.Manufacturing Process Analysis

Patagonia's manufacturing process exemplifies meticulous attention to detail and sustainability at every step. Taking the example of their Down Sweater jacket, we see a thorough process that spans from design to quality control. Notably, Patagonia invests significant time and resources in each stage, ensuring responsible material sourcing and rigorous quality checks.

3.Supply Chain Overview

Patagonia's supply chain is characterized by partnerships with key stakeholders committed to sustainability. From organic cotton farms to responsible down suppliers, the company prioritizes ethical sourcing and environmental responsibility. With a network of fabric mills and production factories worldwide, Patagonia ensures quality and compliance with their standards at every stage of production.

4.Bottlenecks and Capacity

Through analyzing the manufacturing process, it's evident that assembly stands as the bottleneck activity, with a throughput time of 14 weeks and a cycle time of 4 weeks.

Review (1/2)



5. Inventory Management

Patagonia employs a multifaceted approach to manage both raw materials and finished goods. With a focus on social and environmental responsibility, the company ensures compliance with fair labor practices and environmental standards throughout its supply chain. Their 4-Fold Approach to Supply Chain Decisions emphasizes rigorous pre-screening of suppliers to align with Patagonia's high standards.

6. Production Planning and Data Utilization

Patagonia's production planning is deeply integrated with their commitment to sustainability. By assessing their carbon footprint and working closely with suppliers that share their values, the company ensures ethical and transparent operations. Detailed environmental impact assessments and supply chain transparency reports inform their production plans, guiding decisions to minimize harm and prioritize quality.

7. Material Requirement Identification

Patagonia's material requirement identification process reflects their commitment to minimizing socio-ecological harm. Through functional need assessments, responsible sourcing practices, and a focus on recycled and environmentally friendly materials, the company strives to create products that align with their sustainability goals.

8. Quality Management

Quality is paramount for Patagonia, as they combine expert evaluation, thorough testing, and continuous improvement to ensure their materials meet high standards. Through material testing laboratories, traceability requirements, and environmental responsibility programs, Patagonia maintains consistent quality while upholding their commitment to sustainability and ethical practices.

